

Shannon Energy

Sales & Retention Analytics

Power BI demo project - end-to-end build

26,000 acquisitions · 17,000 churn events · 5 sales channels · 2023-2026

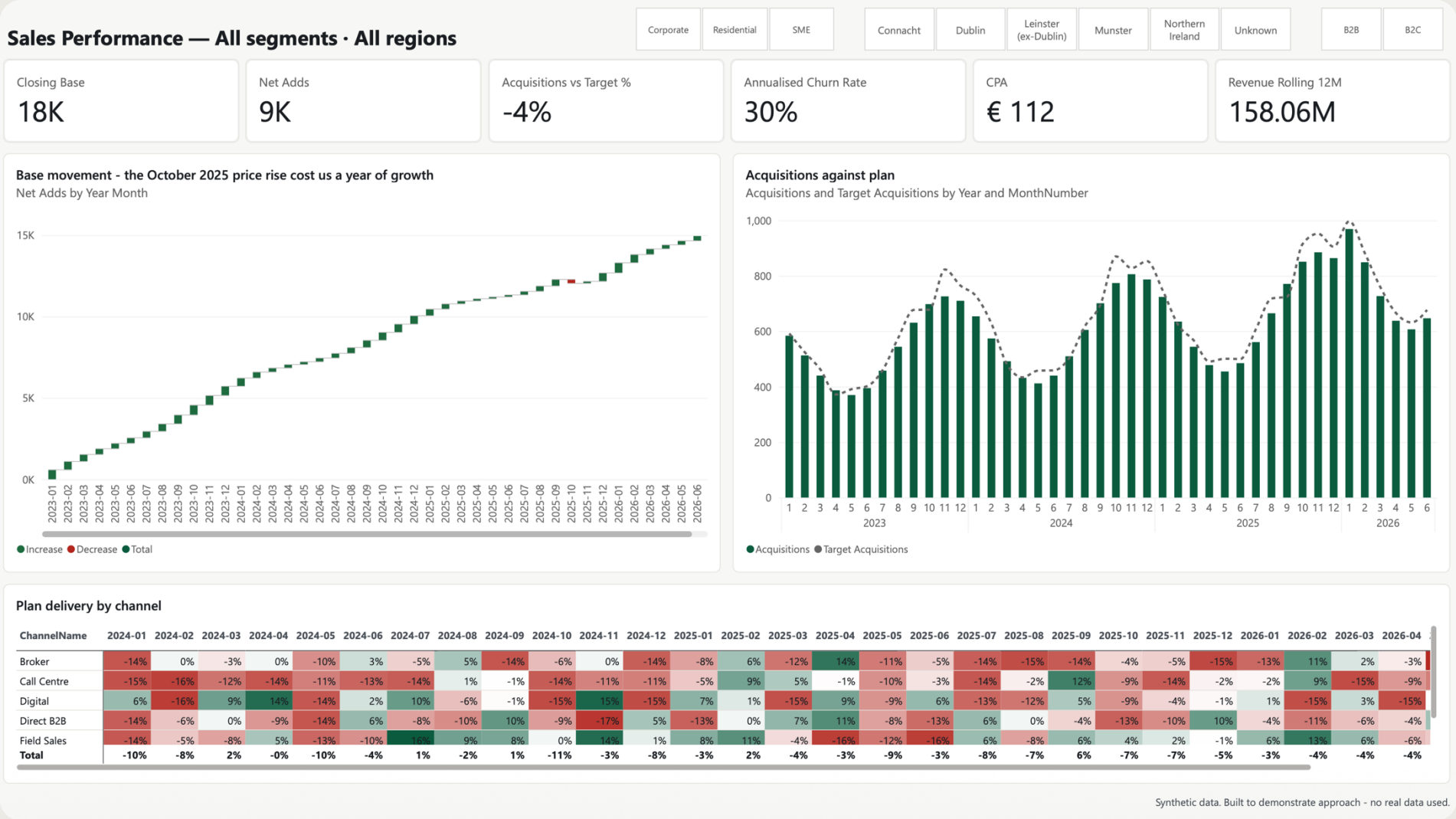
Raw CSV → cleaning in Power Query → star schema → 29 DAX measures → report

Marina Kulmane

Synthetic data. Built to demonstrate method.

The view a sales director opens in the morning

One page: volume, cost, plan vs actual, churn - filtered by segment and region.



Every card is a DAX measure, not a hard-coded number.

The cheapest channel to acquire is the least valuable to keep

B2C: acquisitions, CPA, 12-month retention, cost per retained customer, contribution.

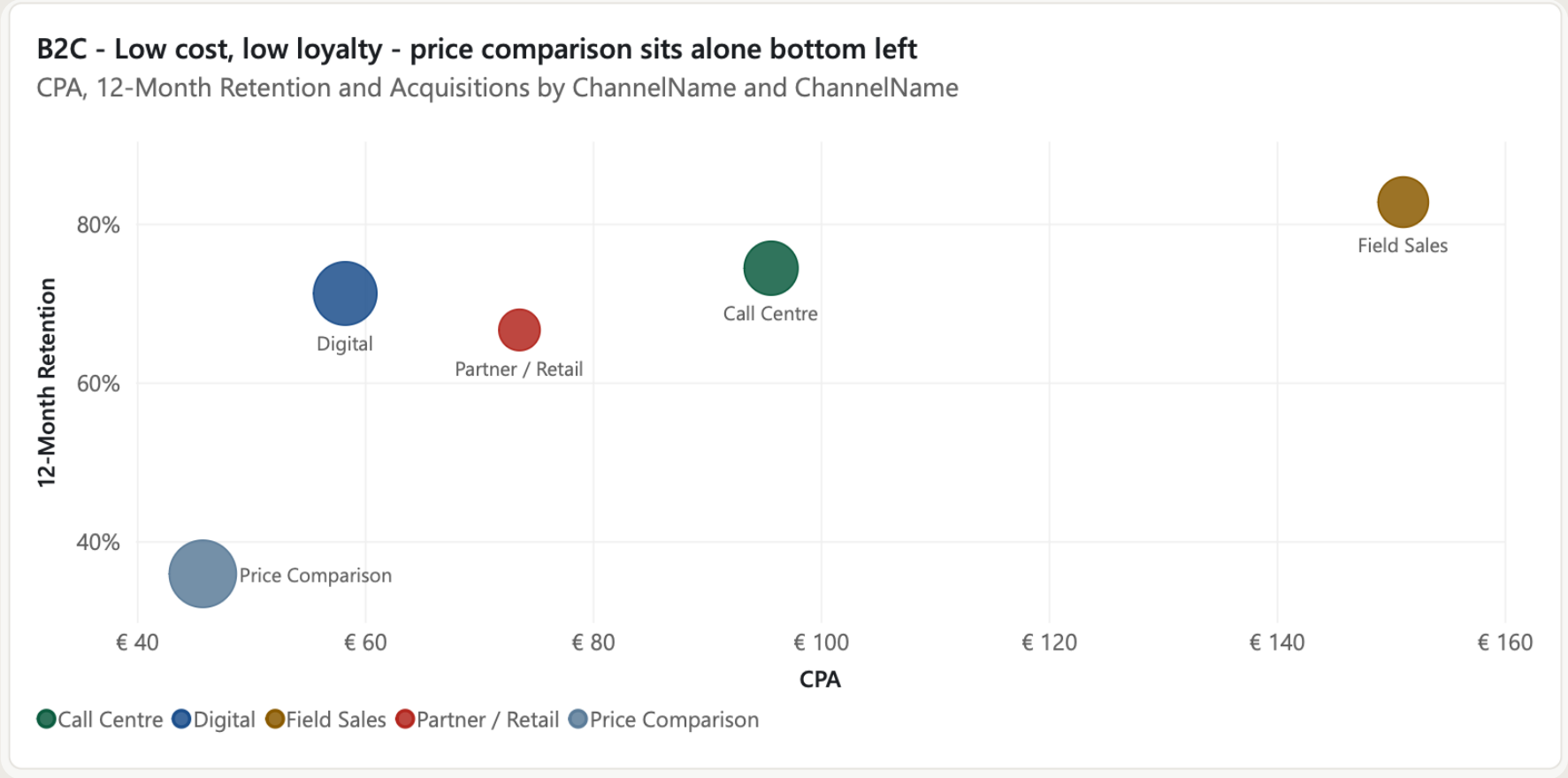
B2C -Cheapest channel to acquire, least valuable to keep

ChannelName	Acquisitions	CPA	12-Month Retention	Cost per Retained Customer	Contribution per Acquisition (24M)
Price Comparison	6,619	€ 46	35.9%	€ 128	€ 130
Digital	5,839	€ 58	71.2%	€ 82	€ 254
Partner / Retail	2,279	€ 74	66.6%	€ 110	€ 218
Call Centre	4,103	€ 96	74.4%	€ 129	€ 239
Field Sales	3,548	€ 151	82.7%	€ 183	€ 229
Total	22,388	€ 78	62.6%	€ 124	€ 212

Price Comparison: CPA €46 - the lowest. Retention 35.9% - the lowest.

Low cost, low loyalty - Price Comparison sits alone

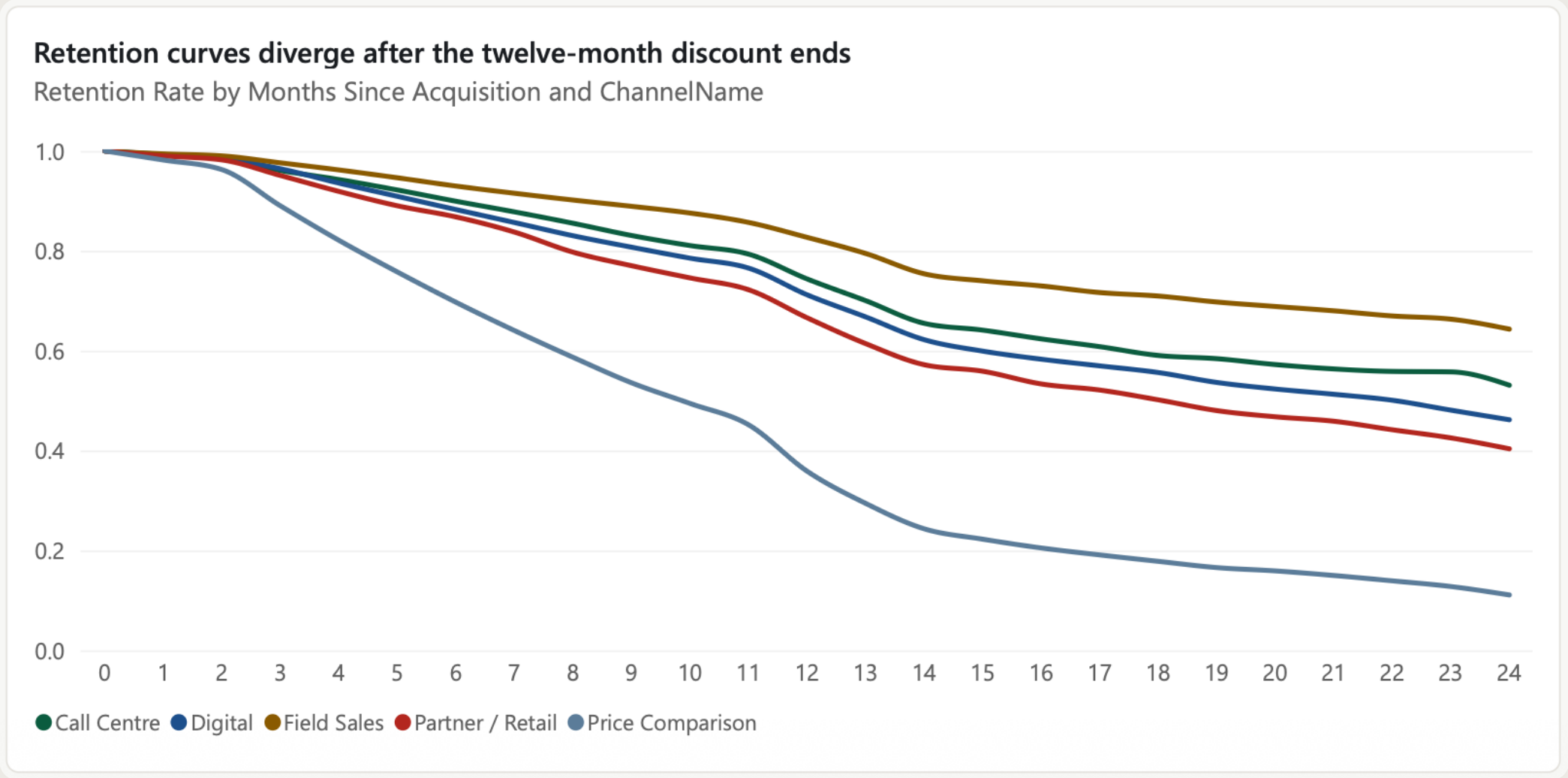
CPA against 12-month retention. Bubble size = acquisitions.



Cost per retained customer turns €46 into €128. Field Sales turns €151 into €183.

The curves diverge when the twelve-month discount ends

Retention curve by months since acquisition, by channel.



Cohorts too young to have reached month N are excluded, not counted as retained.

Blanks are cohorts too young to measure

Cohort matrix: acquisition month * months since acquisition.

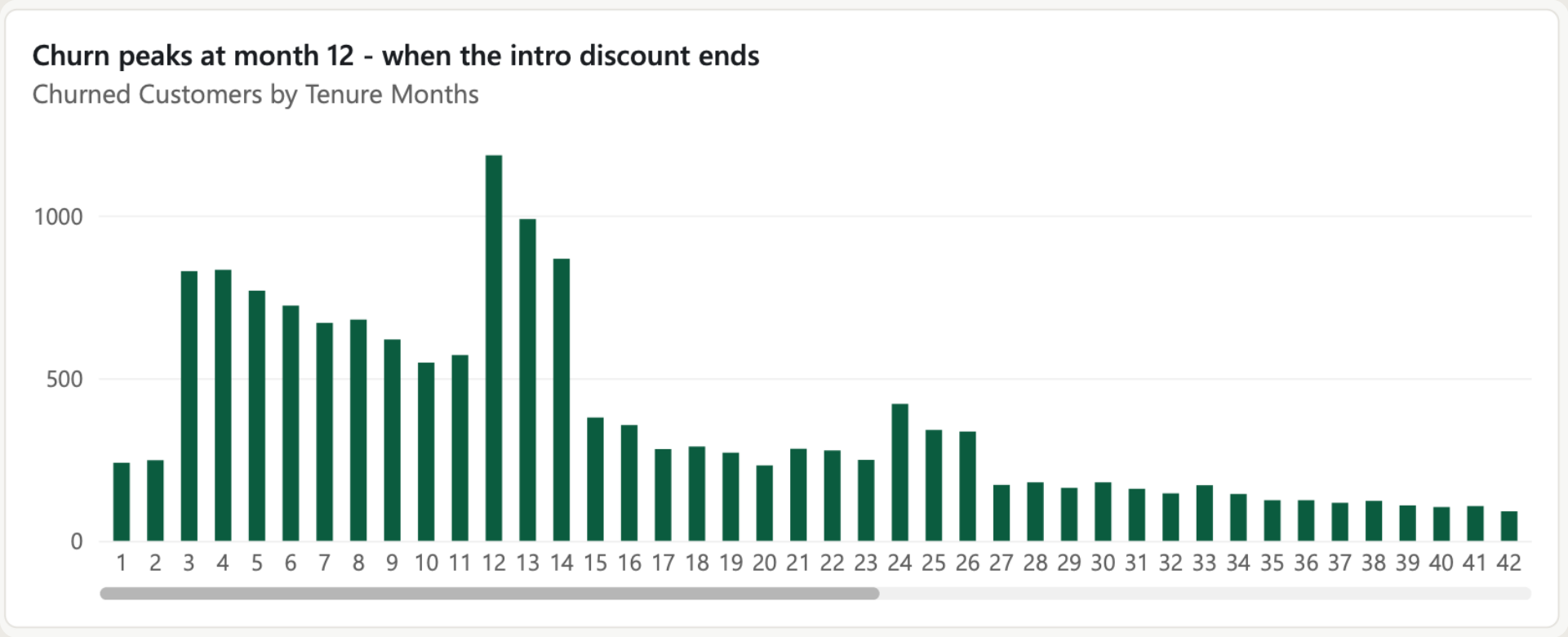
Retention by acquisition cohort - blanks are cohorts too young to measure

Cohort Month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2023-01	1.00	0.99	0.98	0.95	0.92	0.89	0.86	0.83	0.80	0.78	0.76	0.73	0.69	0.65	0.61	0.59	0.58	0.55	0.55	0.54	0.53
2023-02	1.00	0.99	0.98	0.95	0.91	0.88	0.85	0.82	0.79	0.76	0.74	0.73	0.68	0.63	0.60	0.58	0.57	0.55	0.54	0.53	0.51
2023-03	1.00	0.99	0.99	0.95	0.93	0.90	0.87	0.85	0.80	0.77	0.75	0.73	0.68	0.64	0.59	0.59	0.58	0.57	0.54	0.52	0.51
2023-04	1.00	0.98	0.97	0.94	0.90	0.88	0.84	0.81	0.79	0.76	0.73	0.70	0.64	0.58	0.55	0.53	0.51	0.49	0.47	0.45	0.44
2023-05	1.00	0.99	0.99	0.94	0.91	0.88	0.86	0.82	0.78	0.76	0.74	0.72	0.67	0.63	0.57	0.54	0.53	0.51	0.50	0.49	0.49
2023-06	1.00	1.00	0.99	0.97	0.94	0.90	0.88	0.84	0.81	0.78	0.76	0.74	0.71	0.68	0.65	0.64	0.61	0.61	0.59	0.58	0.57
2023-07	1.00	0.99	0.98	0.95	0.91	0.89	0.86	0.83	0.81	0.78	0.76	0.73	0.69	0.65	0.61	0.59	0.57	0.55	0.53	0.52	0.51
2023-08	1.00	0.99	0.99	0.96	0.92	0.90	0.86	0.83	0.80	0.78	0.75	0.74	0.67	0.63	0.58	0.57	0.56	0.55	0.54	0.52	0.51
2023-09	1.00	0.99	0.98	0.94	0.92	0.88	0.85	0.83	0.80	0.77	0.76	0.74	0.68	0.64	0.61	0.60	0.59	0.57	0.56	0.55	0.54
2023-10	1.00	0.99	0.98	0.95	0.92	0.88	0.86	0.83	0.80	0.79	0.76	0.74	0.70	0.65	0.61	0.59	0.58	0.57	0.55	0.54	0.53
2023-11	1.00	0.99	0.99	0.95	0.90	0.88	0.85	0.83	0.80	0.78	0.76	0.74	0.67	0.63	0.60	0.59	0.57	0.56	0.55	0.53	0.52
2023-12	1.00	0.99	0.98	0.95	0.91	0.88	0.86	0.84	0.81	0.78	0.75	0.72	0.66	0.61	0.56	0.54	0.53	0.52	0.50	0.49	0.47
2024-01	1.00	1.00	0.99	0.94	0.91	0.87	0.83	0.80	0.77	0.75	0.73	0.72	0.68	0.63	0.59	0.56	0.55	0.54	0.53	0.52	0.50
Total	1.00	0.99	0.98	0.95	0.91	0.88	0.85	0.82	0.79	0.76	0.74	0.71	0.66	0.62	0.57	0.56	0.54	0.53	0.52	0.50	0.49

A blank is an honest gap in the data. Filling it with 100% is the classic error.

Churn peaks at month 12 - when the intro discount ends

Churned customers by tenure in months.



A pricing question, not a service question - and it is visible before it happens.

Cleaning and security are part of the build

Applied Steps in Power Query · Row-Level Security roles.

Named steps, not "Changed Type1". Anyone can reopen this file and follow it.

Every transformation step, named ->

Twenty spellings of the channel,
currency as text, mixed date formats

Manage security roles

Create new security roles and use filters to define row-level data restrictions.

+ New

B2B Team

B2C Team

Dim_Segment

Rules

Switch to DAX editor

+ New

Select all

Delete

Group

Ungroup

Show data if

All

 of these rules are true

Column	Condition	Value
☐ BusinessType	Equals	"B2C"
+ New		

Save

Close

APPLIED STEPS	
Source	*
Promoted Headers	*
Trim customer IDs	
Remove duplicate acquisition ...	
Remove currency symbol	*
Remove thousands separator	*
Set acquisition cost to decimal	
Handle missing region	*
Remove original region column	
Rename cleaned region	
Map channel to code	*
Filtered Rows	
Normalise channel names	
Set data types	
Merge churn data	*
Expand churn tenure	*
Add churn outcome	
Flag non-churned as -1	*
Add churned flag	*
Inserted Year	*
Changed Type2	*
Inserted Month	*
Inserted Multiplication	*
Inserted Addition	*
Inserted Subtraction	*
Inserted Multiplication1	*
Name months observable to ...	*
Changed Type with Locale	*
Remove helper columns	
Add tenure months	*
Add cohort month	*
Inserted Start of Month	*
Name cohort month	
Add cohort columns	
Final column names	

Two security roles

The B2C and B2B teams don't see each other's numbers

What the numbers say

€46

Price Comparison CPA - the cheapest channel to buy

35.9%

its 12-month retention - the worst of the five

€128

cost per retained customer, once churn is priced in

€130

contribution per acquisition - the lowest of the five